Overview:

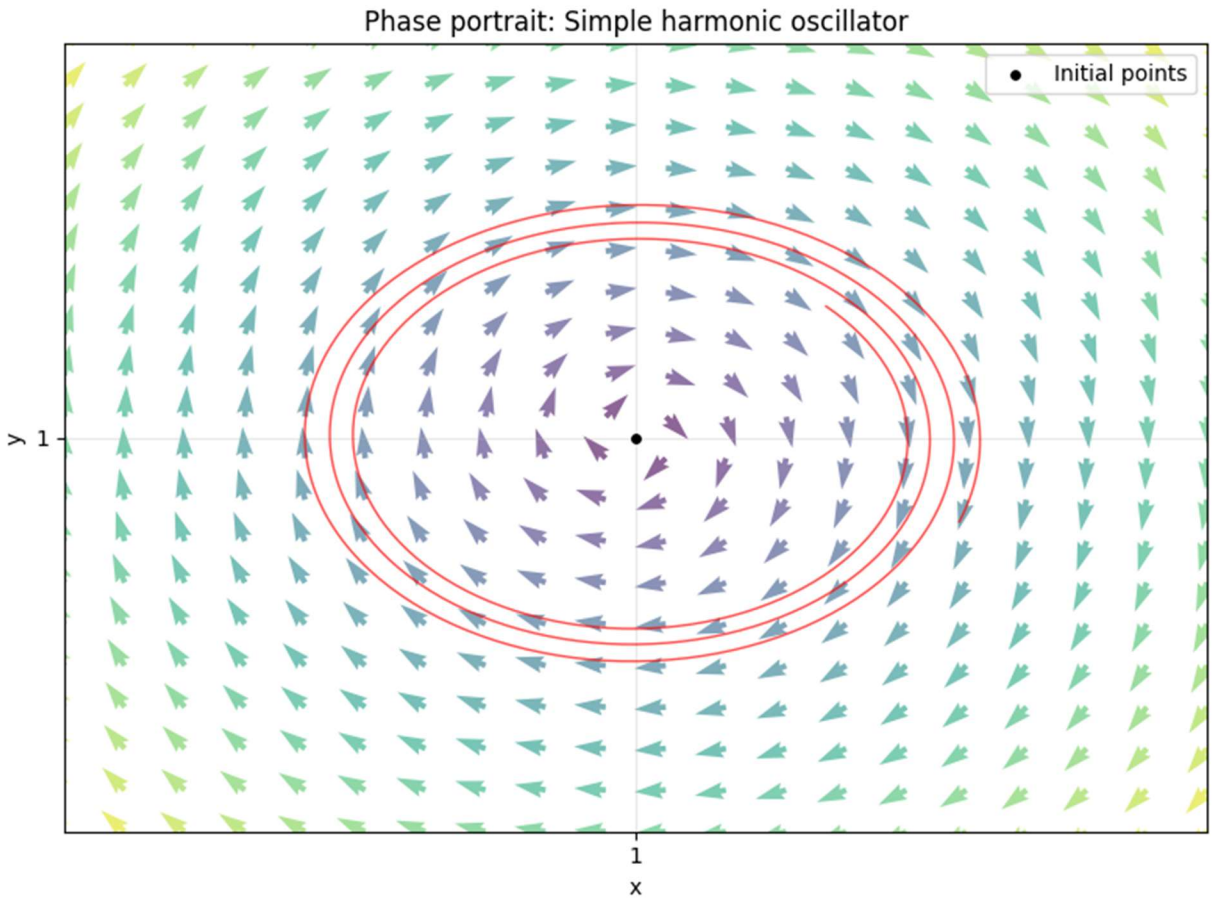
For this project I chose to create a python code that plots the phase diagram of either a harmonic oscillator or of a predator-prey system. It is multiple definitions paired with a main body of code. Each definition if for specific part of the process.

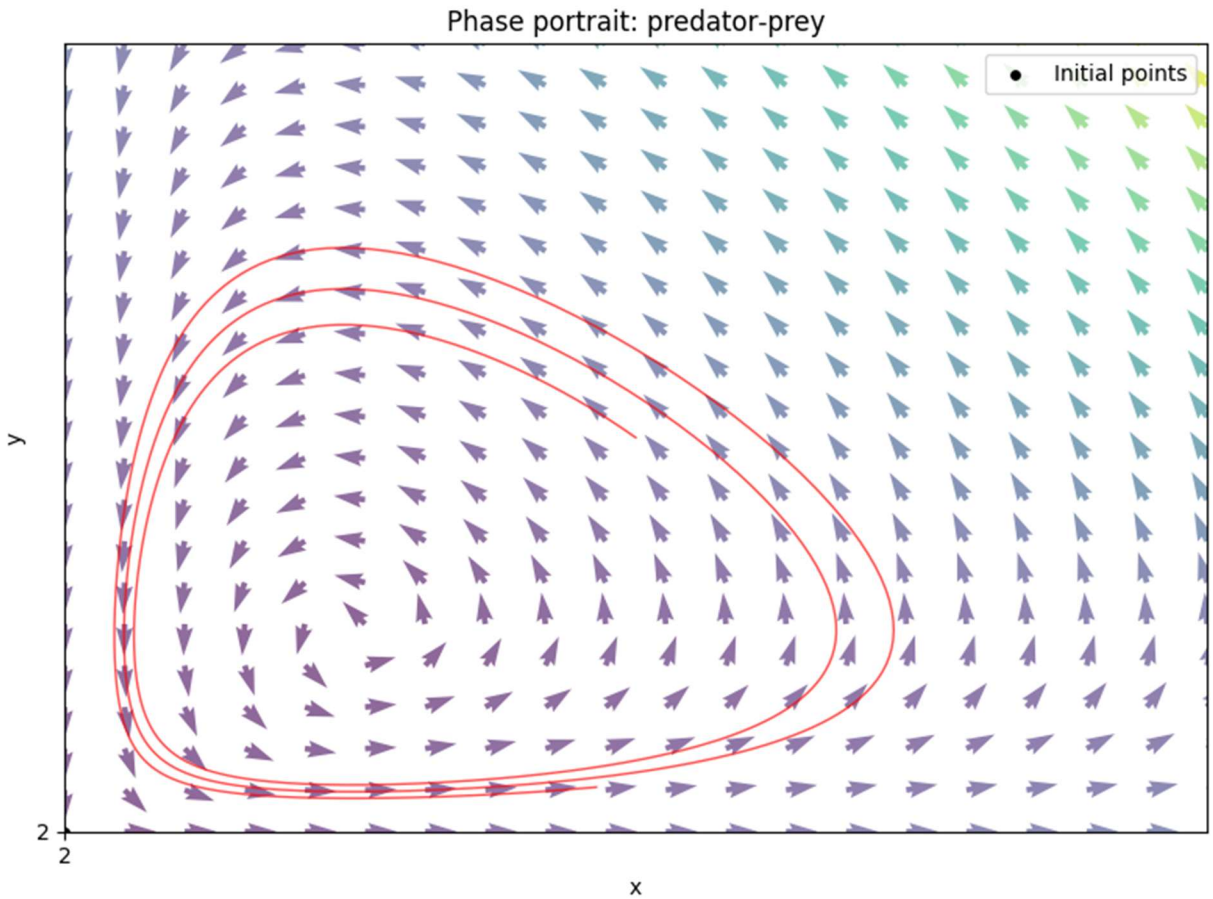
Use:

To use the code, one must insert the function they desire to see into the code itself. This is because I am a mechanical engineering major and couldn't figure out how to do it otherwise. By running the code, you will be prompted to decide which type of diagram you would like to see. Afterwards, you are asked what point you would like to see in the diagram. The diagram is then presented in another window. To run the code again you must clear out the previous diagram.

How it works:

In the main body of code, it asks which plot you would like to see. Based on this answer it calls the main definition and tells it the type of plot. The main definition then creates the grid, then calls on the Euler solver. The Euler solver then does the integration and spits out the y values in an array. This then runs through the equation provided. The trajectories are then plotted. Finally, the diagram appears.


Phase portrait: Simple harmonic oscillator
Initial points
y
1
1
x


Phase portrait: predator-prey
Initial points
y
2
2
x